

Nagarjun Mallesh

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 Expected CTC: ₹30-40 LPA |  Availability: Immediate Joiner |  Status: Based in Bangalore (No relocation needed)

PROFESSIONAL SUMMARY

GenAI Software Engineer with 4+ years of experience building production AI systems and cloud-native distributed architectures. **Based in Bangalore** (relocated from the USA in November 2025), bringing specialized expertise in **AWS Bedrock, RAG pipelines**, and serverless architecture from US-based product companies. Led cross-functional teams delivering 70% performance improvements and 60% cost reductions while maintaining 99.9% uptime and seeking long-term opportunities with Indian product companies or GCCs to apply **GenAI** expertise to solve complex business problems at scale.

Core Expertise: System Design | RAG Implementation | AWS Bedrock & Lambda | Distributed Systems | Team Leadership | Production ML Deployment

EXPERIENCE

Senior Systems Analyst | ML Technologies LLC, Boston, MA | Jul 2025 - Nov 2025 (*Relocated to India for family reasons*)

US-based ML consultancy specializing in emergency services optimization

- Architected and deployed **GenAI-powered vehicle recovery system** using AWS Bedrock (Claude) and OpenSearch vector database, reducing emergency response times by 70% and processing 5,000+ daily requests across 12 metropolitan areas
- Established a technical roadmap for GenAI adoption and **mentored three junior engineers** on RAG best practices, LangChain development, and prompt engineering, accelerating team delivery velocity by 40%
- Designed a **serverless emergency services platform** with API Gateway + Lambda architecture, processing 150,000+ monthly API requests with 99.9% uptime and auto-scaling capabilities
- **Tech Stack:** Python, AWS Bedrock, OpenSearch, Lambda, API Gateway, LangChain, CloudFormation

Backend Software Engineer | Keelworks Foundation, Oak Harbor, WA | Jul 2024 - Jul 2025

US-based non-profit foundation focused on volunteer management technology.

- Led a **cross-functional team of 4 engineers** and QA specialists through the complete SDLC to deliver a production RAG chatbot using AWS Bedrock, serving 10,000+ volunteers with a 92% user satisfaction score
- Reduced AI hallucinations by 45% through the implementation of the **RAGAS evaluation framework** with custom metrics for faithfulness, answer relevancy, and context precision in RAG responses.
- Designed **event-driven serverless architecture** using EventBridge-Lambda workflows for a volunteer coordination platform, achieving 60% cost reduction compared to an EC2-based solution while improving scalability
- Built a **blue-green deployment pipeline** using AWS CodePipeline and CloudFormation, reducing deployment time from 4 hours to 8 minutes with zero-downtime releases
- **Tech Stack:** Python, Node.js, AWS Bedrock, Lambda, EventBridge, DynamoDB, RAGAS, LangChain, Terraform

Software Engineer | Hewlett-Packard Inc, Bangalore, India | Oct 2019 - Jul 2022

Global technology company - BIOS/Firmware automation and management platform development

- Delivered **global automation solution** by developing PowerShell-based BIOS configuration system deployed across HP manufacturing units in 8 countries, transforming 4-day manual processes into 15-minute automated workflows and eliminating 90% of configuration errors
- **Led Infrastructure Automation Project** for AWS-based OS deployment system, reducing provisioning time from 3 days to 4 hours, successfully deployed to 100,000+ client machines globally
- Built a **Management Dashboard Platform** by orchestrating a cross-functional team of 6 engineers and QA to architect a React.js + Node.js web application for real-time BIOS monitoring across 50,000+ devices
- **Tech Stack:** PowerShell, C#, Node.js, React.js, AWS (EC2, S3, Lambda), Python, REST API

Associate Software Engineer | Digital API Craft Pvt Ltd, Bangalore, India | Jul 2019 - Oct 2019

Digital transformation consultancy - Banking sector projects

- Executed **microservices migration** for banking application using Spring Boot and Apache Kafka, handling 50,000+ daily transactions with 99.8% reliability and reducing response latency by 35%
- Built a **domain provisioning automation system** using Golang with Route53 DNS integration, automating the client onboarding process and reducing manual errors by 90%
- **Tech Stack:** Java, Spring Boot, Golang, Apache Kafka, PostgreSQL, AWS Route53

SKILLS

Programming Languages: Python, Java, JavaScript/TypeScript, Node.js, Golang, PowerShell, C#

Cloud & DevOps: AWS (Bedrock, Lambda, API Gateway, EventBridge, S3, EC2, CloudFormation, CloudWatch) | Azure | GCP | Docker | Kubernetes | Terraform | Pulumi | CI/CD (GitHub Actions, CodePipeline)

GenAI & ML: AWS Bedrock (Claude, Nova, Titan models) | LangChain | RAGAS Framework | RAG Pipelines | Prompt Engineering | Vector Databases (OpenSearch, Pinecone) | OpenAI APIs | Hugging Face | Fine-tuning LLMs

Data Technologies: PostgreSQL | MongoDB | Redis | Apache Kafka | OpenSearch | DynamoDB | Data Pipeline Design

Architecture: Microservices | Serverless Architecture | Event-Driven Systems | Distributed Systems | RESTful APIs | System Design

Frameworks: Spring Boot | React.js | Node.js/Express | FastAPI | Flask

EDUCATION

Master of Science in Information Systems | Northeastern University, Boston, USA

Sep 2022 - May 2024 | GPA: 3.6/4.0

Bachelor of Engineering in Information Science | Visvesvaraya Technological University, India

Aug 2015 - Jun 2019 | GPA: 8.0/10.0

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Introduction to Artificial Intelligence (CS50 AI) | Harvard University - 2025**
- **AWS Essential Training for Developers | LinkedIn Learning - 2021**
- **Learning Data Science: Understanding the Basics | LinkedIn Learning - 2021**
- **Learning Git and GitHub | LinkedIn Learning - 2021**
- **Introduction to Big Data | Coursera - 2020 (Completed during HP tenure)**

Academic Coursework: Machine Learning, Deep Learning, Cloud Computing, Distributed Systems (Northeastern University, 2022-2024)

NOTABLE PROJECTS

Multi-Agent Travel Planning System | Personal Project

Engineered production-ready AWS Bedrock-powered platform using Claude 3.5 Sonnet and Nova models with multi-agent orchestration for intelligent travel planning. System handles concurrent user requests via an API Gateway with a Lambda backend, reducing manual planning time by 80% and serving 2,000+ monthly active users with a 95% satisfaction rating and estimated annual savings of \$50K in planning costs.

- **Tech Stack:** Python, AWS Bedrock (Claude/Nova), API Gateway, Lambda, DynamoDB, LangChain
- **Key Features:** Multi-agent workflow, Context-aware recommendations, Real-time API integration

Concurrent Text Normalization Engine | Freelance Project

Built a multi-threaded document processing system leveraging AWS Bedrock LLMs to normalize 10,000+ legal documents simultaneously with 95% accuracy. Implemented producer-consumer pattern with thread pools for optimal resource utilization, reducing processing time from 8 hours to 45 minutes.

- **Tech Stack:** Python, Threading, AWS Bedrock, S3, Lambda, CloudWatch
- **Impact:** 90% reduction in processing time, handling 500GB+ document corpus

AI Travel Matching Platform | Academic Capstone Project

Developed a **Spring Boot web application** with Binary Search Tree algorithms for optimized travel companion matching, integrated with OpenAI GPT-4 for intelligent profile analysis. Served 500+ users during pilot phase with 85% successful match rate.

- **Tech Stack:** Java, Spring Boot, OpenAI API, PostgreSQL, React.js
- **Features:** Real-time matching algorithm, AI-powered compatibility scoring, Secure authentication

TECHNICAL CONTRIBUTIONS & COMMUNITY

Technical Writing: Publish technical articles on Medium covering AI/ML implementation patterns and cloud architecture

Profile: medium.com/@nagarjunmallesh

Code Quality Leadership: Conducted 50+ code reviews across cross-functional teams, establishing best practices for GenAI implementation and serverless architecture